Monitoring and Understanding Notebook-Based Learning with Jupyter

A Dashboard for Student Self-Assessment

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Introduction & Related Work

- **The Problem:** Jupyter notebooks are central to programming education
    - Instructors only see the final code submissions
    - This hides the students' actual learning process

- **The Research Gap:** learning dashboards primarily rely on high-level LMS metrics or focus on instructor-facing analytics
    - A scarcity of fine-grained, student-centered tools that visualize cell-level interactions without increasing cognitive burden.

Current Dashboard Visualizations: Existing systems range from high-level LMS metrics to advanced, coordinated Jupyter extensions for instructors. These include:

- **Real-time classroom monitoring:** Activity heatmaps, student locations, and aggregate performance charts (bar/bubble charts for executions and time spent).
- **Instructor deep-dives:** Cell-level visibility with execution logs and time-slider playback features for teachers to review classroom activity.

Methodology. Research Design

RQ1: How do students evaluate the usability of the learner-facing dashboard?

RQ2: How do students perceive the usefulness and ease of use of the dashboard?

Process

1. Testing the Jupyter Dashboard
2. Responding the Survey
3. Analysis using IBM SPSS Statistics

Methodology. Dashboard Design

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Data Storage (SQLite) *(Passive, unobtrusive, real-time)*

- Users: personal code, role
- Notebooks: title, path, author
- Notebook Sessions: notebook, user, timestamps
- Cells: notebook, cell type, initial content and Jupyter ID
- Active Cells: cell, session, timestamps
- Cell Executions: cell, session, content, timestamps, number of errors and error messages

Methodology. Dashboard Design

Data Visualization

LIVE DEMO!

Results

Sample Characteristics

Table 1. Sample Characteristics ($N = 22$)

Characteristic	Category / Metric	Value
Gender	Female	9 (40.9%)
	Male	11 (50.0%)
	Unspecified	2 (9.1%)
Age (Years)	Mean (SD)	25.00 (2.15)
	Range	21–28
Coding Knowledge	Mean (SD)	4.82 (2.70)
	Range	0–9
Testing Duration (min)	Mean (SD)	17.29 (7.68)
	Range	3–34

Results

Variable Characteristics

Table 2. Descriptive Statistics of Main Study Variables ($N = 22$)

Construct	M	SD	Min	Max
System Usability Scale (SUS)	65.57	17.89	40.00	90.00
TAM Overall	5.36	0.88	3.25	6.75
Perceived Usefulness (PU)	4.93	1.18	3.33	7.00
Perceived Ease of Use (PEOU)	5.78	1.06	3.17	7.00
NASA-TLX (Mental Workload)	8.15	2.92	1.67	12.67

Table 3. Spearman Rank Correlations (ρ) Between Main Study Variables ($N = 22$)

Variable	1	2	3	4	5
1. SUS	—				
2. PU (TAM)	.091	—			
3. PEOU (TAM)	.406	.222	—		
4. TAM Overall	.323	.816*	.706*	—	
5. NASA-TLX	-.099	.239	-.255	.024	—

* $p < .001$, so these correlations are statistically significant.

Discussion

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Usability (RQ 1)

- **System Usability Scale (SUS):** The mean score of 65.57 is slightly below the average benchmark of 68, indicating marginally acceptable usability.
- **Complexity Challenge:** Integrating detailed interaction data, such as cell execution histories and error patterns, requires further UI refinement to reduce perceived complexity.
- **User Feedback:** Students expressed a desire for tighter visual integration, such as showing error lines directly in the code and dashboard.
- **Layout & Cognitive Load:** Vertical scrolling to access all visualizations increases the cognitive burden.
- *Proposed Solution:* Transitioning to a single-page layout to present all summary statistics "at a glance".

Discussion

Usefulness & Ease of Use (RQ 2)

- **Discrepancy in TAM Subscales:**
  - **Perceived Ease of Use (PEOU):** Rated highly (M = 5.78), showing the dashboard is intuitive and easy to navigate
  - **Perceived Usefulness (PU):** Rated more moderately (M = 4.93).
- **Interpretation Barrier:** Students often require time and explicit instructional guidance to effectively interpret analytics feedback and recognize its metacognitive value.
- **"Teacher vs. Student Tool":** Some participants felt the dashboard was more aligned with an instructor's perspective or pure data analysis.
- **Need for Actionable Guidance:** Students want actionable hints on *how* to solve specific errors, rather than just descriptive feedback showing what they did wrong.

Discussion

Mental Workload (NASA-TLX)

- **Low Subjective Workload:** The NASA-TLX yielded a mean overall score of 8.15 on a 20-point scale.
- **Below Global Average:** When transformed to a 100-point scale (~40.75), this sits comfortably below the global median of 43.8 for user-experience meta-analyses.
- **Key Success:** The dashboard acts as a low-workload addition to the programming environment and successfully avoids the "cognitive burden" trap.
- **Focus Maintained:** It provides transparent execution histories without distracting learners from their primary goal of solving Python exercises.

Limitations & Future Work

Limitations

- Sample size (only 22 participants)
- Avg interaction duration (17 mins)

Future Work

- Comparing students with & without dashboard
- Multiple learning units to see improvement
- Cell execution durations to study optimization
- Database of verified solutions
- Teacher dashboard
    - Compare students
    - Overall problems

Conclusion

- **Core Objective:** The dashboard successfully makes the iterative, trial-and-error nature of programming transparent to learners.
- **Key Findings:** The concept is highly viable; it is perceived as very easy to use and effectively avoids adding cognitive hurdles.
- **Pedagogical Needs:** Unlocking the tool's full instructional value requires longitudinal engagement, pedagogical guidance, and concrete problem-solving hints.
- **Future Impact:** Process-oriented analytics can sustainably bridge the gap between evaluating final code correctness and understanding individual learning paths.

Thank you for your attention!

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